DEMAND ESTIMATION · US CAR MARKET 1971–1990

# What Car Buyers Will Pay

Estimating willingness to pay, price elasticities and marginal costs with the Berry–Levinsohn–Pakes (BLP) method in R

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**$3,980**
Value of air conditioning to buyers

**$4,280**
Value of a larger car (+1 SD in size)

**−1.8**
Average own-price elasticity

**67% → 21%**
Cars wrongly price-insensitive, OLS → IV

Dollar values in 1983 dollars.

## 1 Summary

Using only public sales, price and spec data, this project estimates what US car buyers in 1971–1990 were willing to pay for each car feature. It replicates the logit benchmarks of Berry, Levinsohn and Pakes (1995), the standard method for estimating demand for differentiated products.

- **Willingness to pay (WTP):** the dollar value buyers place on a feature, recovered from how sales respond to features versus price.
- **Price elasticities:** how much a car's sales fall when its price rises. The average car loses about 1.8% of sales for a 1% price increase.
- **Marginal costs:** backed out from prices and demand alone, with no cost data needed.
- **Why the method matters:** a naive regression says buyers *dislike* air conditioning and treats 67% of cars as price-insensitive. Correcting for price endogeneity fixes both.

## 2 The data

The classic BLP US car dataset: **2,217 car models** sold across **20 yearly markets** (1971–1990) by **26 manufacturers**. It ships with the R package `BLPestimatoR` and has no missing values.

| Variable | Meaning | Average |
|---|---|---|
| price | Price, $1,000s (1983 dollars) | 11.8 |
| hpwt | Horsepower per 10 lbs of weight (performance) | 0.39 |
| air | 1 if the car has air conditioning | 24% of cars |
| mpg | Fuel economy, tens of miles per gallon | 2.1 (21 mpg) |
| space | Length × width, 1,000 sq. in. (size) | 1.31 |
| share | Car's sales ÷ all US households | 0.1% |
| cdid / firmid | Market (year) / manufacturer | — |

Each year about **89% of households buy no new car**. This "outside option" is part of the model: a price rise can push buyers to another car or out of the market altogether. The data is a set of 20 yearly snapshots; the same model gets a new id each year, so cars are compared within and across years rather than tracked over time.

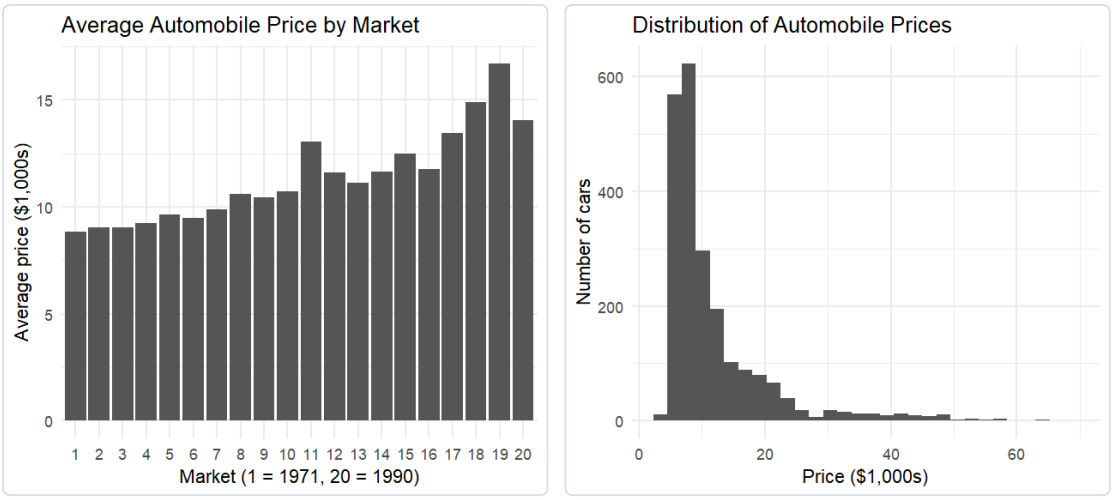



Figure 1. Average price by market rises from about $9k in 1971 to $14-17k by the late 1980s (left); prices are right-skewed, with most cars at $5-12k and a long luxury tail (right).

## 3 The method in plain words

The model compares cars' sales with their prices and features: if bigger cars sell more at the same price, buyers value size. Dividing a feature's value by how much buyers dislike paying gives its dollar worth.

1. **Turn shares into a regression.** In the logit demand model (Berry, 1994), each car's log share minus the log of the "buy nothing" share is a straight line in price and features:

$$\ln(s_j) - \ln(s_0) = -\alpha\, p_j + \beta_1 \mathrm{hpwt}_j + \beta_2 \mathrm{air}_j + \beta_3 \mathrm{mpg}_j + \beta_4 \mathrm{space}_j + \xi_j$$

2. **Fix the price problem (endogeneity).** $\xi$ is quality the data doesn't record, such as reliability or styling. Better cars cost more *and* sell more, so a plain regression (OLS) mistakes quality for "buyers don't mind high prices".

3. **Use instruments.** Following BLP (1995), price is predicted from the features of the same firm's *other* cars. A firm with many similar cars of its own charges higher markups, but a buyer's liking for one car doesn't depend on that. Only this predicted part of price is used (IV regression).

4. **Add year fixed effects.** These absorb anything common to all cars in a year, such as recessions or interest rates. Standard errors are clustered by manufacturer.

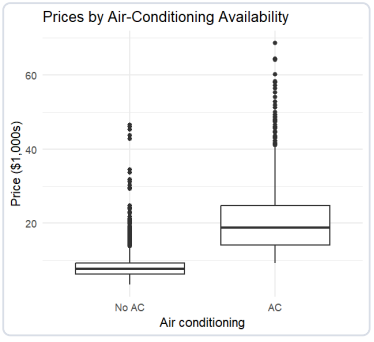



Figure 2. Cars with air conditioning are far pricier, partly because AC came on better cars: the endogeneity problem in one picture.

## 4 Results: correcting for endogeneity matters

Once price is instrumented, buyers look about **75% more price-sensitive**: the price coefficient moves from −0.089 to −0.155. The OLS estimate matches the original BLP paper's (−0.088).

| Coefficient | OLS | OLS + year FE | IV + year FE (main) |
|---|---|---|---|
| **price** | **−0.089*** ** | **−0.089*** ** | **−0.155*** |
| hpwt (performance) | 0.526 | 0.654 | 2.655 |
| air (AC) | 0.016 | −0.036 | 0.616 |
| mpg (fuel economy) | 0.501** | 0.527* | 0.335 |
| space (size) | 2.740*** | 2.867*** | 2.792*** |
| $R^2$ | 0.39 | 0.41 | 0.33 |

2,217 observations per model. * p < 0.05, ** p < 0.01, *** p < 0.001; standard errors clustered by manufacturer.

### Instrument checks

| Test | Result | What it means |
|---|---|---|
| First-stage F | 69.4 (needs > 10) | Instruments are strong predictors of price |
| Wu–Hausman | p < 0.001 | Price is endogenous, so IV is needed |
| Sargan | p < 0.001 | Instruments don't fully agree; a known weakness of the simple logit |

## 5 Willingness to pay for car features

Buyers valued air conditioning at about **$3,980** and a one-standard-deviation larger car at about **$4,280** (1983 dollars). A feature's WTP is its coefficient divided by the price coefficient:

$$\mathrm{WTP}_k = \beta_k \,/\, \alpha$$

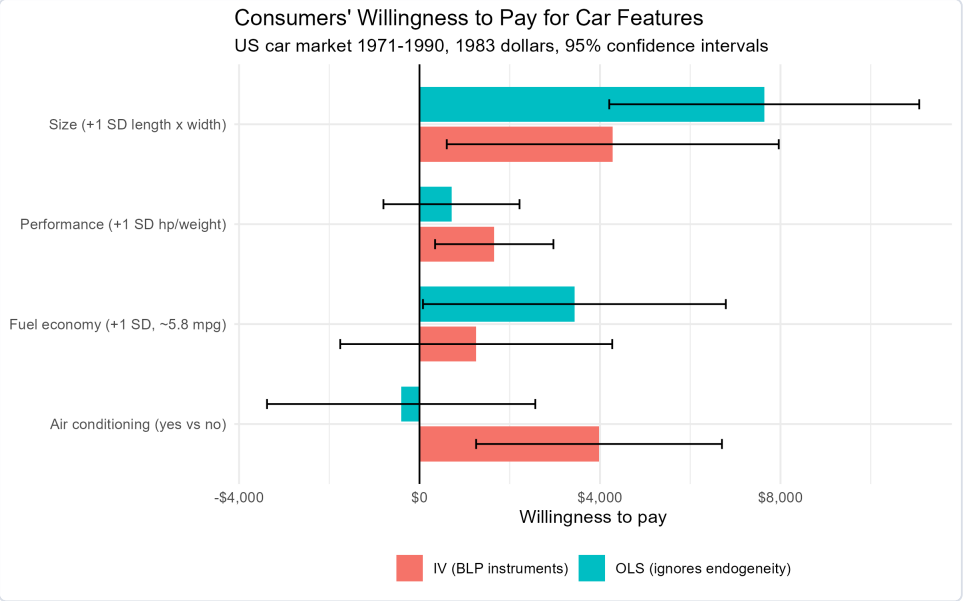



Figure 3. Willingness to pay by feature, OLS vs IV, with 95% confidence intervals.

| Feature (IV estimate) | WTP | 95% range | Clearly above zero? |
|---|---|---|---|
| Size, +1 SD | $4,283 | $600 to $7,960 | Yes |
| Air conditioning, yes vs no | $3,979 | $1,260 to $6,700 | Yes |
| Performance, +1 SD hp/weight | $1,657 | $350 to $2,970 | Yes |
| Fuel economy, +1 SD (~5.8 mpg) | $1,257 | −$1,750 to $4,260 | No |

**Why the method matters here**
OLS says buyers *dislike* AC (−$407). AC came mostly on expensive, high-quality cars, so OLS confuses the AC premium with the price penalty. IV removes that confusion.

**Business use**
These numbers price a feature. A carmaker could charge up to about $4,000 more for AC before the average buyer walks away, or judge whether adding a feature is worth its cost.

## 6 Price elasticities

With IV, the average car loses **1.8% of its sales for a 1% price rise**, and only 21% of cars look price-insensitive, down from 67% with OLS. In the logit model:

$$\varepsilon_{jj} = -\alpha \, p_j \, (1 - s_j)$$

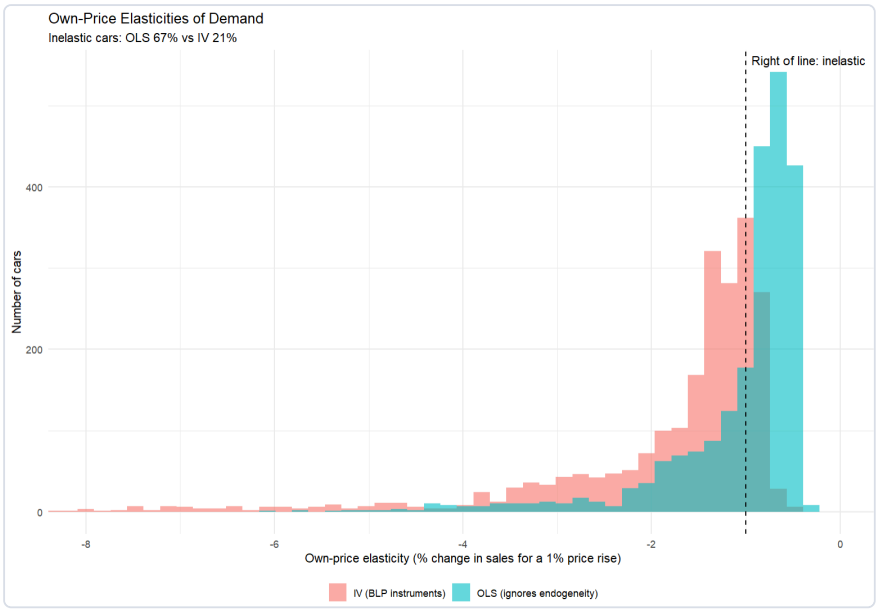



Figure 4. Own-price elasticities, OLS vs IV. Right of the dashed line, demand is inelastic.

An elasticity between 0 and −1 means a price rise would *increase* revenue. A profit-maximising firm would keep raising price, so such cars shouldn't exist; fewer cars in that zone means a more believable model.

| Price band | Cars | Average elasticity (IV) |
|---|---|---|
| Under $8k | 903 | −0.99 |
| $8k–$12k | 675 | −1.47 |
| $12k–$20k | 378 | −2.38 |
| Over $20k | 261 | −4.81 |

**Business use**
Elasticities tell a pricing team how many sales a price change costs. For a car at −2, a 5% discount lifts sales by about 10%.

**Caution**
The rise with price is built into the logit formula, not a finding. Real luxury buyers are richer and usually *less* price-sensitive.

## 7 Reverse-engineering markups and marginal costs

Without any cost data, the model puts the typical markup at about **$6,500 per car**, and marginal cost at price minus that amount. The logic: if firms set prices to maximise profit, the price-sensitivity estimate tells us how far above cost they can go. For a firm selling several cars (logit, Bertrand pricing):

$$p_j - mc_j = 1 \;/\; [\; \alpha\,(1 - S_f)\;]$$

$S_f$ is the firm's combined market share in that year.

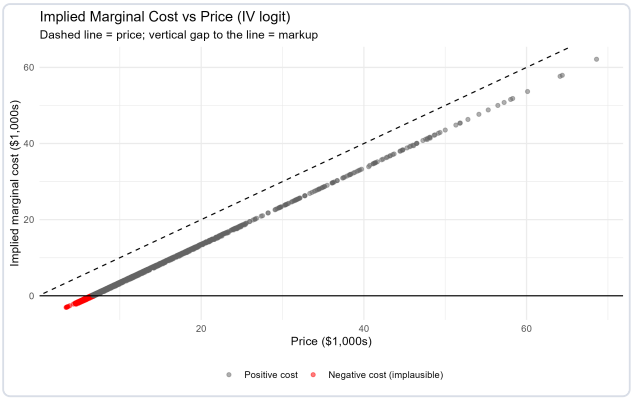



Figure 5. Implied marginal cost vs price (IV logit). The vertical gap to the dashed line is the markup.

**Business use**
A firm can estimate rivals' costs and margins from public prices and sales, which helps with pricing, merger analysis and competitive strategy.

**What the chart reveals**
In the simple logit, every car of a firm gets almost the same dollar markup. Cheap cars priced under about $6,500 then come out with *negative* costs (red dots, 23% of cars), which is impossible. This is the main reason the full BLP model is needed.

## 8 Limitations and next step

The logit model gets the method working but is too rigid for final pricing numbers. The next step is the **full BLP random-coefficients model**, which lets buyers differ in their tastes and income.

- **Everyone has the same tastes.** So elasticities rise mechanically with price, and markups are a flat dollar amount (hence the negative costs).
- **Unrealistic substitution.** If a Mercedes gets pricier, the logit sends its buyers to a cheap Chevette as readily as to a BMW.
- **Sargan test rejects.** The instruments don't fully agree, likely because the logit is too simple.

- **Historical data.** Results describe the US market in 1971–1990, in 1983 dollars, not today's buyers.

## Sources

Berry, S., Levinsohn, J. and Pakes, A. (1995). *Automobile Prices in Market Equilibrium.* Econometrica, 63(4).

Berry, S. (1994). *Estimating Discrete-Choice Models of Product Differentiation.* RAND Journal of Economics, 25(2).

Data: `productData_cars` in the R package BLPestimatoR. Analysis code: `code base.R`.